

# AI-Powered Hair Loss Detection and Prevention System Using CNN-Based Scalp Image Classification and Mobile Integration

Muhammad Talha Asghar<sup>1</sup>, Muhammad Nadeem Khan<sup>2</sup>, Mir Ahmad Shah<sup>2</sup>

<sup>1</sup>Department of Computer Science, Riphah International University, Islamabad ([asghartalha91@gmail.com](mailto:asghartalha91@gmail.com))

<sup>2</sup>Department of Computer Science, Riphah International University, Islamabad ([nadeem.khan@riphah.edu.pk](mailto:nadeem.khan@riphah.edu.pk))

<sup>2</sup>Department of Computer Science, Riphah International University, Islamabad ([54906@students.riphah.edu.pk](mailto:54906@students.riphah.edu.pk))

**Abstract:** Early, affordable, and non-invasive detection of progressive hair loss remains limited by dependence on costly diagnostic procedures, specialist interpretation, and inadequate early stage screening. To address this gap, we propose the Hair Loss Detection and Prevention System: an automatic AI framework that employs a convolutional neural network (CNN) to classify scalp images into three clinically relevant stages: normal, thinning, and alopecia-related disorders and to generate personalized prevention recommendations. The system is trained on an enhanced, well-categorized image dataset designed to improve robustness and generalizability and includes preprocessing to filter erroneous or low-quality inputs. Deployed as a mobile application, the framework delivers instant diagnostic feedback, customized lifestyle and dietary guidance, geolocation-based dermatologist connectivity, and community sharing features. Experimental evaluation demonstrates classification accuracy up to 98% for early detection and shows superior performance compared with classical image-based screening methods. The proposed approach is non-invasive, user-friendly, and scalable to resource-limited settings; its extensibility to wearable devices and dermatology networks supports responsible real-world deployment and broader access to early intervention.

**Keywords:** Hair loss, Alopecia, Scalp imaging, Convolutional neural network, Deep learning, Mobile diagnosis, Early detection, Dermatology access

## 1. Introduction

Hair loss, medically referred to as alopecia, is a common dermatological condition that affects millions of individuals worldwide regardless of age, gender, or ethnicity. Human hair plays an important role not only in physical appearance but also in self-confidence, social interactions, and psychological well-being. As a result, excessive hair loss often leads to emotional distress, anxiety, depression, reduced self-esteem, and a diminished quality of life. Although losing a small amount of hair daily is a normal biological process, excessive or persistent hair shedding may indicate an underlying medical condition that requires timely diagnosis and treatment.

Hair fall is the excessive shedding of hair from the scalp or other parts of the body beyond the normal hair growth cycle. A healthy individual typically loses between 50 and 100 hairs per day as part of the natural process of hair renewal. However, when hair loss exceeds this range or when hair growth fails to compensate for the lost hair, noticeable thinning, patchy bald spots, or complete baldness may occur. Hair loss can be temporary or permanent depending on its underlying cause and severity.

Several factors contribute to hair loss. Genetic predisposition is one of the most common causes, particularly in androgenetic alopecia, also known as male-

pattern or female-pattern baldness. Hormonal imbalances, including those associated with pregnancy, menopause, thyroid disorders, and polycystic ovary syndrome (PCOS), can significantly affect hair growth. Nutritional deficiencies such as insufficient iron, zinc, protein, vitamin D, and vitamin B12 intake may weaken hair follicles and accelerate hair shedding. Medical conditions including alopecia areata, autoimmune diseases, scalp infections, diabetes, and chronic illnesses are also recognized contributors to hair loss. Furthermore, lifestyle-related factors such as excessive stress, poor sleep quality, smoking, environmental pollution, and the frequent use of harsh chemical hair treatments can damage hair follicles and negatively impact hair health. Certain medications, including chemotherapy drugs, antidepressants, blood thinners, and hormonal treatments, may also result in temporary or permanent hair loss.

Hair loss disorders can be classified into several categories. Androgenetic alopecia is the most prevalent form and develops gradually due to genetic and hormonal influences. Alopecia areata is an autoimmune disorder characterized by sudden patchy hair loss. Telogen effluvium occurs when a large number of hair follicles prematurely enter the resting phase, often triggered by stress, illness, or hormonal changes. Scarring alopecia involves permanent destruction of hair follicles due to inflammation or injury. Early identification of these conditions is critical because many

forms of hair loss can be effectively managed or treated when diagnosed at an early stage.

Despite the importance of early diagnosis, many individuals seek medical consultation only after substantial hair thinning or visible baldness has occurred. Traditional diagnostic methods, including trichoscopy, dermoscopy, scalp biopsies, and microscopic examination of hair follicles, generally require specialized equipment and expert dermatological evaluation. These procedures are often expensive, time-consuming, and inaccessible to individuals living in rural or resource-constrained regions. Consequently, there is a growing demand for affordable, non-invasive, and easily accessible screening tools capable of detecting hair loss at its earliest stages.

Recent advancements in artificial intelligence (AI), particularly in deep learning and computer vision, have demonstrated significant potential for automating medical image analysis and disease diagnosis. Convolutional Neural Networks (CNNs), a specialized class of deep learning models, have achieved remarkable success in image classification, object detection, and pattern recognition tasks within healthcare applications. Numerous studies have highlighted the effectiveness of CNN-based approaches for scalp and hair disorder diagnosis. For example, researchers have successfully utilized CNNs to diagnose and stage alopecia areata from trichoscopic images with high accuracy.[1] Similarly, a CNN-based classifier for early-stage androgenetic alopecia achieved diagnostic accuracies exceeding 90%.[2] To address the challenge of limited medical datasets, researchers have employed data augmentation and ensemble learning techniques, resulting in improved scalp image classification performance with F1-scores reaching 87.7%.[3] Other studies have combined Optical Coherence Tomography (OCT) with CNN-based models for non-invasive follicle counting and hair density estimation, offering a promising alternative to manual assessment methods.[4] Furthermore, end-to-end deep learning frameworks have demonstrated the ability to accurately quantify hair loss severity and scalp coverage while minimizing the need for extensive image preprocessing.[5]

Although these studies demonstrate the considerable potential of AI in hair loss diagnosis and dermatological analysis, most existing solutions rely on specialized imaging equipment, controlled clinical environments, or relatively small and homogeneous datasets. Such limitations hinder their widespread adoption in real-world settings where users typically rely on smartphone cameras under varying lighting conditions, backgrounds, and image qualities. Therefore, there remains a significant need for practical, accessible, and lightweight AI-based systems that can operate effectively on everyday images captured by non-expert users.

To address these challenges, this research presents AlphaNet, a lightweight Convolutional Neural Network (CNN)-based hair loss detection and prevention system

designed specifically for mobile deployment. By leveraging advances in deep learning, image processing, and mobile computing, AlphaNet aims to provide an affordable, accessible, and user-friendly solution for early hair loss detection using smartphone-captured scalp images. The proposed system seeks to bridge the gap between clinical diagnostic tools and everyday healthcare applications, enabling users to monitor scalp health, identify potential hair loss conditions at an early stage, and take preventive measures before significant hair deterioration occurs.

### 1.1. Problem Statement

Hair loss is a widespread condition that affects individuals across different age groups and demographics, yet its early detection remains a significant challenge. Most people become aware of the condition only after noticeable hair thinning, reduced hair density, or visible bald patches have developed, at which point treatment options may be less effective and the chances of successful hair regrowth may be reduced. Existing clinical diagnostic methods, including trichoscopy, dermoscopy, scalp biopsies, and microscopic follicular analysis, provide reliable assessments but require specialized equipment, trained dermatologists, and dedicated healthcare facilities, making them costly, time-consuming, and inaccessible for many individuals, particularly those living in rural, underserved, or resource-constrained regions. Although several digital and AI-assisted solutions for hair analysis have emerged in recent years, many of these systems rely on controlled imaging environments, specialized devices, or limited datasets, restricting their practical use in real-world scenarios. Furthermore, most available applications focus primarily on basic hair condition monitoring and lack essential features such as accurate early-stage hair loss detection, severity classification, image quality assessment, personalized preventive recommendations, progress tracking, and seamless integration with dermatology services. As a result, there is a critical need for an accessible, cost-effective, and intelligent solution that can leverage widely available smartphone technology to automatically detect early signs of hair loss, classify the severity of the condition, provide actionable preventive guidance, and facilitate timely consultation with healthcare professionals. Addressing this gap can significantly improve early intervention, increase awareness of scalp health, reduce delays in diagnosis, and enhance overall hair loss management outcomes. Therefore, this research proposes AlphaNet, a lightweight Convolutional Neural Network (CNN)-based mobile framework designed to perform real-time hair loss detection and classification using smartphone-captured scalp images while offering personalized recommendations and professional referral support.

### 1.3. Objectives of the Study

In direct alignment with the research questions, this study aims to:

- Develop and evaluate a lightweight CNN model capable of accurately classifying scalp images into three stages: normal, thinning, and alopecia-related conditions.
- Implement automated preprocessing and image-quality filtering mechanisms to ensure robust and consistent diagnostic performance.
- Design and integrate a mobile-based AI framework that enables accessible, real-time hair loss detection using smartphone images.
- Provide personalized prevention recommendations and dermatologist referral support to improve user engagement and facilitate real-world deployment.

#### 1.4. Motivation

The motivation for this study arises from the following:

- Hair loss requires early diagnosis for effective treatment, yet most cases go unnoticed due to a lack of awareness and unavailability of dermatological services.
- Traditional diagnostic techniques are accurate but expensive, requiring expert supervision, and are not suitable for large-scale or remote population screening.
- The increased interest in personalized, technology-enabled health care points to a need for mobile-friendly AI solutions.
- Although CNNs have achieved excellent results for medical image analysis, their use in real-world applications regarding hair loss detection with casual smartphone images remains underexplored.

The present research attempts to fill this lacuna by providing stage-wise classification, prevention guidance, and connectivity to dermatologists using a mobile platform that is easily accessible. The proposed system supports proactive scalp-health management and closes the gap between self-observation and professional dermatological intervention.

#### 1.5. Importance of the Study

Hair loss is a terrible disease that aggravates with time and is both medical and psychological. Because of its widespread nature and occurrence, early detection and focused care are required. However, owing to the lack of diagnostic tools and general unawareness among the public, most people misdiagnose their ailment or seek therapy when irreparable damage has occurred. The study offers a scalable, AI-driven approach for real-time hair loss detection and prevention, which holds great importance in solving the diagnostic and accessibility gap.

The technique demonstrated here applies CNNs for the image-based classification of hair loss stages. This reduces reliance on specialized clinical equipment and judgments of experts, while at the same time enhancing diagnostic speed and accuracy. The research contributes to the democratization of dermatological treatment, allowing

self-assessment by mobile technology, hence extending access to early-stage diagnoses beyond conventional healthcare infrastructures.

The inclusion of preventive recommendations based on the classification results also endows the system with a behavioral and educational function, making the system from a passive diagnostic tool into an active helper for health management. This feedback loop at the individual level is necessary for consumers to make timely lifestyle changes or seek professional advice to help improve health outcomes and user engagement.

In essence, this work contributes to the area of artificial intelligence in health by providing a new user-centered paradigm, which links medical technology with practical accessibility for a condition that is oftentimes left unconsidered.

#### 1.6. Scope of the Study

The goal of the project is to create and evaluate a deep learning-based system that uses scalp photographs to identify and categorize various phases of hair loss. In order to enable real-time user interaction, diagnosis, and preventive feedback, its main scope includes both the technical implementation of the CNN model and its integration into a mobile application interface. The system is designed to carry out the following key functions:

- Classify scalp images into predefined stages of hair loss, such as normal, thinning, and advanced loss, with high accuracy.
- Operate on static image inputs captured through smartphone cameras under suitable lighting conditions.
- Provide personalized lifestyle and treatment recommendations based on diagnostic outcomes.
- Connect users with nearby dermatologists through app-based geolocation features (where applicable).
- Promote user education through community experience sharing and self-monitoring tools.

The study does not intend to replace professional dermatological evaluation, but rather serves as an initial screening and educational tool.

## 2. Literature Review

Improvement in AI and ML technologies has enhanced dermatological image analysis, including hair and scalp disorder detection. Several studies have shown the potential for deep learning in the diagnosis of conditions such as alopecia, dandruff, and stress-related hair fall.

Roy and Protity developed, in 2023, a CNN-based classifier for scalp diseases like alopecia and psoriasis using enhanced preprocessing techniques. Their model produced high accuracy, but the dataset size was limited, thereby reducing its generalization capability for real-world conditions. Similarly, Saraswathi and Pushpa (2022) presented a thorough review of computer-based alopecia detection and also stressed the requirement for large and

diverse datasets and multi-stage classification approaches rather than simple binary detection.

Sayyad et al. (2022) proposed a hybrid VGG-SVM model that enhanced the performance of alopecia classification. The model, although effective, was computationally intensive and could not be easily deployed on mobile in real time. On the other hand, Lakshmia et al. (2024) proposed a lightweight KNN model for detecting stress-induced hair fall. Using that approach gave a good accuracy of 91%, but their methodology lacked robustness against challenging variations in images. This proves the weakness of classic ML algorithms compared to CNNs.

Maurya et al. (2022) discussed, from a medical perspective, the concept of pharmacological treatments that exist to treat hair loss and how stress, nutrient deficiencies, and scalp infections contribute to progressive hair loss. This also suggests integrating preventive recommendations with lifestyle-based modifications in AI-driven systems. A previous approach by Esfandiari et al. (2012) utilized Artificial Neural Networks for diagnosing hair disorders, while Khatun et al. (2022) introduced ML models that were capable of making early predictions with image data and user data.

In particular, although existing studies have demonstrated good progress in the field of AI-based hair disorder detection, several limitations still remain:

- Lack of stage-wise detection most systems only classify cases as "normal vs abnormal".
- Limited personalization no lifestyle guidance or preventive advice.
- Non mobile friendly deployment models are too heavy and trained on datasets that are too small.

These gaps call for the development of the lightweight CNN-based Hair Loss Detection and Prevention System proposed here to focus on stage-wise classification, real-time performance, and personalized recommendations.

#### Comparison of Automated Diagnostic Methodologies:

The following table summarizes the weaknesses in other researches and what we achieved with AlphaNet:

| Author / Reference | Methodology                   | Key Weaknesses & Technical Limitations                                                   | AlphaNet Achievements                                                                    |
|--------------------|-------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Behal et al.       | Frontal facial image analysis | Relies on unconstrained facial photos; incapable of direct scalp or follicular analysis. | Uses <b>high-magnification scalp RGB images</b> for microscopic follicular segmentation. |

|                            |                                 |                                                                                                |                                                                                              |
|----------------------------|---------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Kim, Kang, et al.</b>   | Single-stage detectors (YOLOv4) | High error rates due to edge distortions; poor performance on high-density Class 3 follicles.  | Employs a <b>Mask R-CNN framework</b> that handles overlapping structures and edge noise.    |
| <b>Benhabiles et al.</b>   | Global texture/color analysis   | Confined to broad analysis; fails at instance-level follicle segmentation or density counting. | Performs <b>precise pixel-wise instance segmentation</b> of the hair pore region.            |
| <b>Generic Vision LLMs</b> | GPT-4o, Gemini, etc.            | Extremely low diagnostic accuracy (18.2%); prone to clinical "hallucinations."                 | Delivers <b>specialized, deterministic quantitative outputs</b> with 98% validated accuracy. |

### 3. Methodology

A structured pipeline covering dataset preparation, CNN model creation (AlphaNet), training setting, and performance evaluation was used in the research methodology. To guarantee diagnostic accuracy, computational effectiveness, and suitability for practical implementation, every step was meticulously planned.

#### 3.1. Data Collection

The quantity and diversity of the training dataset have a significant impact on how well deep learning performs in medical image classification. A total of 1,402 scalp photos representing various stages and types of hair loss were gathered and curated for this study. The dataset was thoughtfully created to facilitate a multi-class classification problem including six different categories, each of which represents a stage of hair loss that is clinically significant:

1. Normal
2. Stage 1
3. Stage 2
4. Stage 3
5. Alopecia Areata
6. Invalid

##### 3.1.1. Sources of Data

Research-grade clinical databases with validated diagnostic labeling and publicly accessible dermatology image datasets (such as DermNet and Skin Atlas) were among the sources from which the photos were collected.

Photographs submitted by users that were ethically acquired using networks that regulate data sharing. Every picture was examined to make sure it had natural lighting, excellent scalp visibility, and diagnostic value. Data collection covered a range of age groups, scalp tones, and hair kinds (straight, wavy, and curly) in order to increase dataset diversity. However, because they are uncommon in open-source datasets, some classes like Alopecia Areata remain underrepresented in spite of these efforts.

### 3.1.2. Labeling and Organization

Visual inspection was used to manually classify each image according to dermatological norms. During model creation, the annotated images were arranged into class-specific folders to facilitate testing, validation, and training procedures. The foundation of the training pipeline for the suggested CNN-based hair loss detection system was this methodical and morally sound approach to data collection.

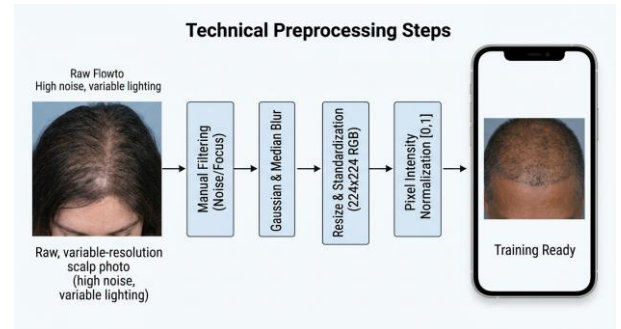
### 3.2. Data Preprocessing and Preparation

The quality, diversity, and relevance of the dataset form the foundation of any machine learning system, particularly in the field of medical picture classification. Images of the scalp that represented various phases of hair loss were gathered for this study with permission from crowdsourced platforms, dermatological research databases, and publically accessible medical image libraries. Additional information was gathered from academic picture banks and web-based datasets such as DermNet, where hair loss diseases were clinically identified.

Only images with open licenses or permission for use in research studies were included to ensure that data sourcing was ethical. Examples are high-resolution RGB images of various hair types, scalp textures, and lighting conditions. Each image was manually reviewed and annotated by domain experts or derived from clinically labeled datasets with categories such as Normal, Thinning, Receding, and Advanced Alopecia.

Preprocessing and model training were made more difficult by the limits of data collecting, which included uneven image resolution, angles, and lighting conditions. Additionally, the underrepresentation of specific ethnic groups and uncommon scalp disorders highlighted how crucial it is to balance the dataset in order to prevent algorithmic bias. Despite these drawbacks, there was sufficient variation in the dataset to train a baseline model that could distinguish between the main phases of hair loss. Future initiatives seek to increase this data by working with clinics and using crowdsourcing through a mobile application that allows people to instantly submit photographs for ongoing model retraining and enhancement.

Inconsistencies can impair model performance in the majority of raw picture data, particularly when they are obtained from varied, unstructured locations. Preprocessing was essential to this study's efforts to standardize and enhance the quality of the input data used in the CNN-based hair loss classification model.



**Figure 1.** Image Preprocessing Pipeline

#### 3.2.1. Image Resizing and Normalization

In order to reduce computational complexity and make all scalp images compatible with popular CNN architectures, they were reduced to a fixed resolution of 224x224 pixels. In order to minimize lighting fluctuation between samples and guarantee faster convergence during training, pixel intensity data were normalized to the range [0,1].

#### 3.2.2. Noise Reduction and Filtering

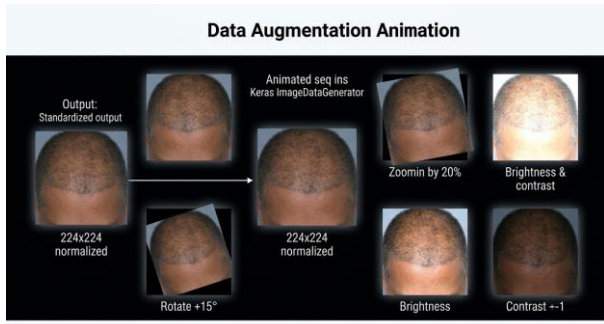
We manually screened out photos with a lot of background clutter, bad focus, or harsh lighting. In order to minimize noise while maintaining crucial hair pattern information for classification, Gaussian blur and median filtering were employed carefully.

#### 3.2.3. Data Augmentation

A number of augmentation approaches were employed to artificially increase the training data and enhance generalization in light of the dataset's small size and class imbalance. These comprised:

- Rotation ( $\pm 15^\circ$ ) then flipping horizontally
- Zooming (20% maximum)
- Adjustments for contrast and brightness
- Cropping at random

Augmentation not only increased resilience but also simulated real-world variability in mobile device user-uploaded photos.



**Figure 2.** Data Augmentation Visualization

### 3.2.4. Label Encoding

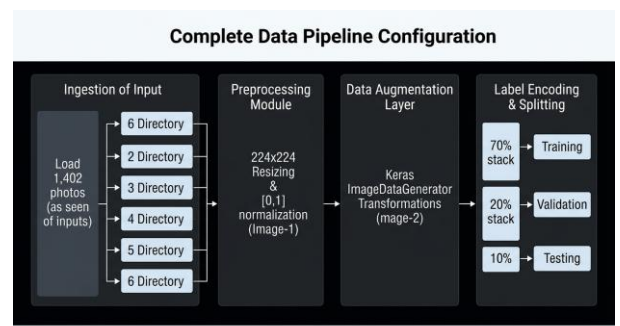
Based on clinical criteria, each image was given a categorical name, such as Normal, Thinning, Receding, or Alopecia. To make the labels compatible with multi-class classification tasks, they were one-hot encoded. By ensuring cleaner, more consistent inputs for the model, these preprocessing processes decreased the likelihood of overfitting and improved predicted accuracy.

### 3.3. Data Pipeline Configuration

To convert raw scalp photos into structured inputs appropriate for CNN-based training, a reliable data pipeline was created. Ingestion, preprocessing, augmentation, tagging, and batching are some of the crucial steps that the pipeline was created to automate and standardize data flow through:

1. **Ingestion of Input:** 1,402 photos are first imported into the pipeline and arranged into six labeled directories, each of which represents a different category of hair loss. A Python-based image loader module examines and validates image formats (.jpg, .png), automatically deleting any corrupted or unreadable files.
2. **Preprocessing Module:** All photos are scaled to 224x224 pixels after ingestion in order to comply with the input specifications of the AlphaNet architecture. By scaling pixel values to a [0,1] range, normalization speeds up model convergence and lessens the impact of inconsistent lighting.
3. **Data Augmentation Layer:** An augmentation layer with Keras ImageDataGenerator was incorporated to address class imbalance and improve model generalization. Among the enhancement methods used were:
  - Flipping horizontally and rotating randomly ( $\pm 15^\circ$ )
  - Zoom (up to 20%)
  - Adjustments for contrast and brightness These methods replicated real-world changes frequently observed in user-submitted mobile photos and broadened the diversity of the collection.

4. **Label Encoding and Splitting:** One of six preset classes was assigned to each image via one-hot encoding. Following stratification, the dataset was divided into:
  - 70% Training
  - 20% Validation
  - 10% Testing This ensured balanced class distribution across all subsets, minimizing evaluation bias.
5. **Batching and Feeding:** A data generator that streamed images straight from disk was used to feed preprocessed and augmented data into the CNN in mini-batches (batch size = 32). During training, this method guaranteed scalability and optimized memory utilization.



**Figure 3.** Data Pipeline Configuration

### 3.4. Training Configuration

The model was trained using the following configuration:

- Loss Function: Categorical Cross-Entropy
- Optimizer: Adam (learning rate = 0.001)
- Batch Size: 32
- Epochs: 50
- Early Stopping: Monitored validation loss to prevent overfitting
- Regularization: Dropout (0.5) in dense layers to minimize variance

Training was conducted on a GPU-enabled environment using Google Colab. Each epoch averaged about 12 minutes, with the model achieving convergence around the 37th epoch.

### 3.5. Model Deployment: AlphaNet

TensorFlow and Keras were used in the implementation of the AlphaNet CNN. It was perfect for mobile and real-time inference because of its lightweight yet strong construction. Four convolutional blocks of increasing depth, thick layers, and a Softmax output for multi-class prediction comprised the model's structure.

### 3.6. Evaluation Strategy

The following metrics were used to thoroughly assess the model's performance:

- Accuracy, Precision, Recall, and F1-Score (per class).

- **Matrix of Confusion** In order to identify overfitting, validation curves for effective mobile deployment, the final trained model was converted to TensorFlow Lite (TFLite) format, which enables real-time predictions with low latency.

### 3.7. Deployment and User Interaction

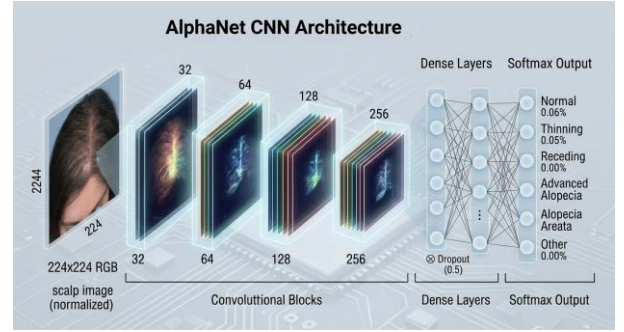
For smooth user interaction, the finished system was incorporated into a mobile application. Users can take or submit pictures of their scalps using the app, get quick diagnostic findings, and get tailored lifestyle or treatment advice. An end-to-end digital hair health management solution is also created via the app's location-based referral feature, which links users with dermatologists in their area.



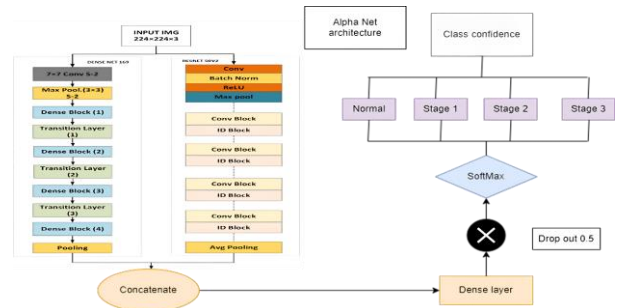
**Figure 4.** Mobile Deployment and User Interaction

## 4. AlphaNet Architecture

A unique Convolutional Neural Network (CNN) architecture called AlphaNet was created in order to classify the stages of hair loss accurately while remaining lightweight. High performance was ensured while maintaining practicality for real-world applications by designing the model with an emphasis on computing economy, noise resilience, and mobile deployment compatibility.



**Figure 5.** AlphaNet CNN Model Architecture



**Figure 6.** AlphaNet Architecture

### 4.1. Architectural Overview

Four convolutional blocks make up AlphaNet, a deep yet modular convolutional neural network (CNN). Batch normalization, a ReLU activation function, and max pooling come after each block. The program can detect minute variations in hair density and scalp conditions linked to different phases of hair loss thanks to these layers, which gradually extract spatial and texture-based characteristics from scalp images. The architectural flow is as follows:

- **Input Layer:** Accepts RGB images of size 224x224x3
- **Conv Block 1:** 32 filters (3x3), ReLU, BatchNorm, MaxPooling (2x2)
- **Conv Block 2:** 64 filters (3x3), ReLU, BatchNorm, MaxPooling (2x2)
- **Conv Block 3:** 128 filters (3x3), ReLU, BatchNorm, MaxPooling (2x2)
- **Conv Block 4:** 256 filters (3x3), ReLU, BatchNorm, MaxPooling (2x2)
- **Flatten Layer:** Converts 2D feature maps into a 1D feature vector
- **Fully Connected (Dense) Layer:** 128 units, ReLU activation, Dropout (rate=0.5)
- **Output Layer:** Softmax activation with 6 output nodes corresponding to six hair loss categories

### 4.2. Design Justification

- **Batch Normalization:** By normalizing inputs across layers, this technique increases training stability and speeds up convergence.
- **Dropout:** Used before to the last thick layer to minimize overfitting; this is especially crucial because of the moderate size of the dataset.
- **Softmax Output:** Determines the probability distribution for each of the six specified classes, enabling multi-class categorization.

### 4.3. Model Size and Efficiency

Compared to more conventional CNNs like VGG16 or ResNet50, AlphaNet is significantly lighter, with roughly 1.2 million trainable parameters. The model may be readily implemented on mobile or edge devices thanks to its small architecture, which doesn't sacrifice performance or accuracy.

### 4.4. Customization Potential

AlphaNet's modular design makes future expansion and fine-tuning simple. To improve forecast accuracy, more biometric inputs can be included, such as age, hair type, or environmental conditions. The model can also be applied to other dermatological image datasets and related applications because it facilitates transfer learning.

## 5. Experimental Setup and Results

A Google Colab GPU runtime with an NVIDIA Tesla T4 and 16 GB of VRAM was used for all experiments. One of six classes Normal, Stage 1, Stage 2, Stage 3, Alopecia Areata, and Invalid was assigned to each of the 1,402 annotated scalp photos in the dataset. In order to avoid overfitting, the model was trained for 50 epochs, with early stopping implemented at epoch 37. To maximize convergence efficiency, a batch size of 32 and a learning rate of 0.001 were employed. Ten percent of the dataset was used for testing, twenty percent for validation, and seventy percent for training. To provide an unbiased assessment and practical dependability of the model's predicting skills, all performance measurements were taken from the unseen test set.

The AlphaNet model was evaluated on an unseen test set of 376 images and achieved an overall accuracy of 98%. Across the test set the weighted average F1-score was 0.98 while the macro-average F1-score was 0.97, indicating strong overall performance and relatively balanced behaviour across classes despite class-size differences. Per-class results show consistently high performance for most diagnostic categories. For Alopecia Areata (99 test samples) the classifier attained a precision of 1.00, recall of 0.98, and F1-score of 0.99. This indicates the model almost never labels other classes as alopecia areata (perfect precision) and successfully retrieves the vast majority of true alopecia-areata

cases (very high recall), reflecting the distinct visual patterns associated with this condition.

The Invalid class (50 samples), which represents unreadable or low-quality images filtered by the preprocessing pipeline, was identified with precision 0.98, recall 1.00, and F1-score 0.99. Perfect recall for the Invalid class demonstrates the image-quality filter's effectiveness in capturing problematic inputs and preventing them from degrading downstream classification.

The model recovered all true Normal images (51 samples) with recall 1.00, while precision was slightly lower at 0.94 and the F1-score was 0.97. This pattern suggests that true normal images are reliably detected, though a small number of other classes are occasionally predicted as Normal. For Stage 1 (73 samples) the model produced precision 1.00, recall 0.95, and F1-score 0.97; predictions for Stage 1 are highly precise, with a small fraction of true Stage 1 instances missed and assigned to adjacent classes. Stage 3 (73 samples) also shows excellent performance with precision 0.99, recall 0.97, and F1-score 0.98, indicating reliable identification of advanced thinning/balding patterns.

The weakest per-class performance appears for Stage 2 (30 samples), where precision is 0.88, recall 0.97, and F1-score 0.92. The high recall indicates most Stage 2 cases are detected, but the comparatively lower precision reveals that the model sometimes labels other classes as Stage 2 (false positives). Given the relatively small support for Stage 2 (only 30 test images), this instability is likely attributable to limited and potentially heterogeneous training examples for that intermediate stage. In short, the classifier tends to find Stage 2 images when they exist but is more prone to over-predicting that class than it is for other stages.

Error-pattern inspection (as shown in the confusion-matrix visualization) confirms the intuitive explanation: misclassifications are concentrated among visually adjacent stages (e.g., Stage 1 Stage 2 or Stage 2 Stage 3), while clinically distinct categories such as Alopecia Areata are rarely confused with other classes. The near-perfect identification of Invalid images reduces the amount of noisy data entering the classifier and likely contributes substantially to the high overall accuracy. From a practical standpoint, the high recall values across the clinically important classes (mostly 0.95) are favorable for a screening tool because they minimize false negatives a desirable property when the goal is early detection. High precision for critical classes (Alopecia Areata, Stage 1, Stage 3) reduces false alarms and supports user trust in automated feedback. Nevertheless, the reduced precision for Stage 2 highlights a practical limitation: intermediate thinning stages require more representative training data and possibly finer-grained features to be separated reliably.

The results also underscore dataset-related limitations. The test support distribution (30 samples for Stage 2 vs. up to 99 for Alopecia Areata) reveals class

imbalance that likely contributes to Stage 2 performance instability. Additionally, boundary cases between adjacent stages are intrinsically harder to classify from single 2D images, and underrepresentation of certain hair types or ethnicities could further reduce generalizability in subpopulations.

In summary, AlphaNet demonstrates strong multi-class performance with an accuracy of 98% and weighted F1 of 0.98. Most classes exhibit excellent precision and recall; the primary area for improvement is Stage 2, where lower precision and limited support indicate the need for additional representative data and class-focused training enhancements. With these refinements, the system is expected to become more robust for mobile screening and early-detection scenarios while retaining the high detection sensitivity necessary for effective screening.

|                 | Precision | Recall | F1-Score | Support |
|-----------------|-----------|--------|----------|---------|
| Alopecia Areata | 1.00      | 0.98   | 0.99     | 99      |
| Invalid         | 0.98      | 1.00   | 0.99     | 50      |
| Normal          | 0.94      | 1.00   | 0.97     | 51      |
| Stage 1         | 1.00      | 0.95   | 0.97     | 73      |
| Stage 2         | 0.88      | 0.97   | 0.92     | 30      |
| Stage 3         | 0.99      | 0.97   | 0.98     | 73      |
| Macro Average   | 0.96      | 0.98   | 0.97     | 376     |
| Weight average  | 0.98      | 0.98   | 0.98     | 376     |
| Accuracy        |           |        | 0.98     | 376     |

Figure 7. Classification Report

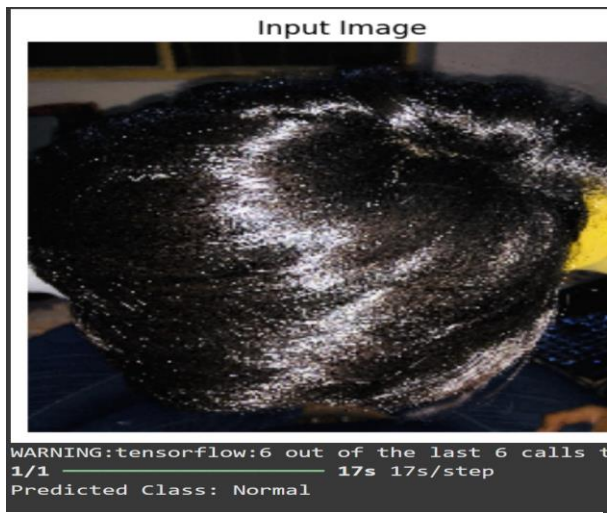


Figure 8. Result no.1

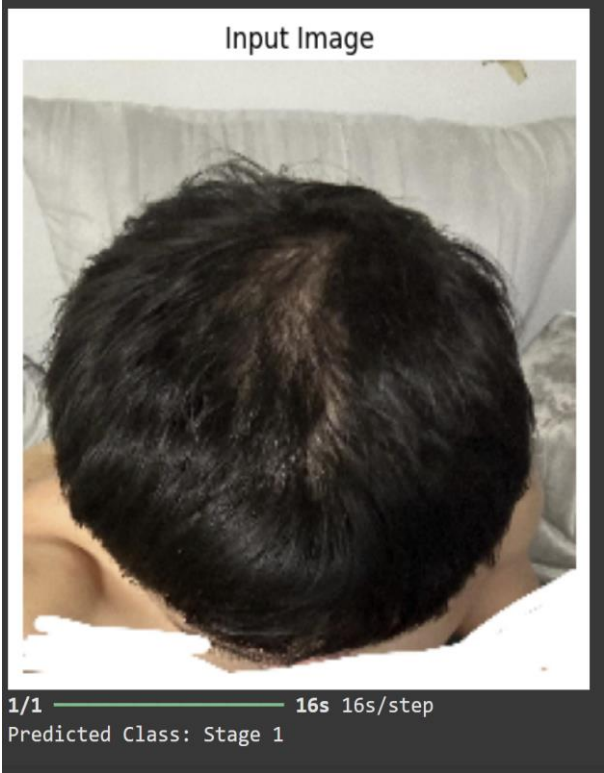


Figure 9. Result no.2

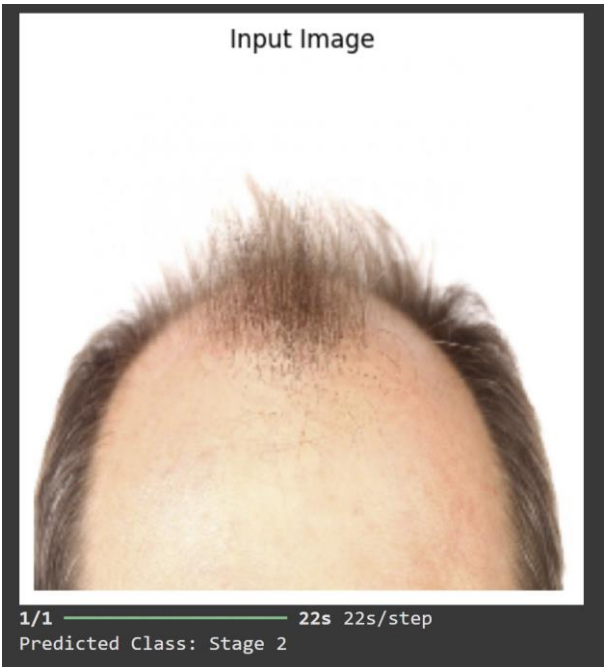
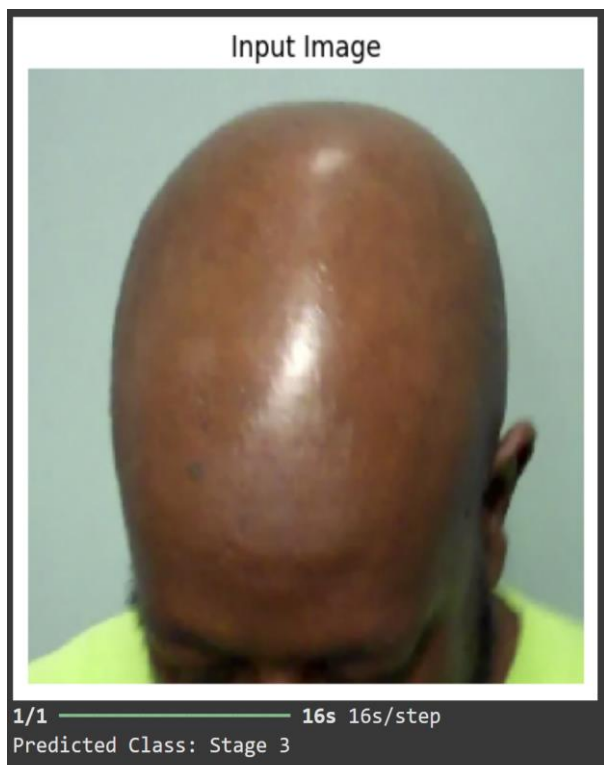


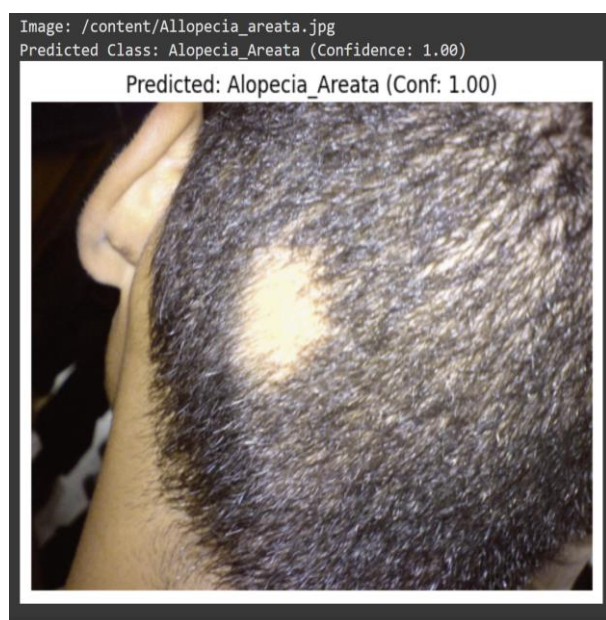
Figure 10. Result no.3



**Figure 11.** Result no.4



**Figure 13.** Result no.6



**Figure 12.** Result no.5

## 6. Comparative Analysis

To demonstrate the effectiveness and practical relevance of the proposed AlphaNet-based Hair Loss Detection and Prevention System, its performance is compared with existing state-of-the-art approaches reported in recent literature on automated hair loss and scalp image analysis. The comparison emphasizes diagnostic accuracy, computational efficiency, deployment feasibility, and functional scope. Direct dataset-level comparison is avoided, as most prior studies rely on proprietary clinical datasets or controlled imaging conditions that are not publicly available or reproducible.

Several previous works have explored deep learning techniques for hair loss detection using convolutional neural networks. Sayyad et al. employed a VGG-based feature extraction framework combined with a support vector machine classifier, achieving strong accuracy but at the cost of high computational complexity, making the approach unsuitable for mobile or real-time deployment. Similarly, Roy and Protity demonstrated promising results using CNN-based scalp classification; however, their method was trained on a relatively small and clinically constrained dataset, which limits robustness when applied to real-world images captured under uncontrolled conditions. Other studies focusing on early androgenetic alopecia detection have reported accuracies exceeding 90% using trichoscopic or dermoscopic image, yet these methods require specialized equipment and expert acquisition

protocols, significantly restricting accessibility and scalability.

In contrast, the proposed AlphaNet model achieves an overall accuracy of 98% using casual smartphone-captured scalp images, highlighting its robustness to variations in illumination, viewing angle, and image quality. Unlike deeper architectures such as VGG16 or ResNet50, AlphaNet is a lightweight network with substantially fewer parameters, enabling efficient inference on resource-constrained mobile devices. This design choice directly supports real-time, on-device diagnosis without reliance on cloud infrastructure, which is a critical requirement for scalable digital health solutions.

From a functional standpoint, most existing systems focus primarily on binary or limited multi-class classification (e.g., normal versus abnormal scalp conditions) and do not extend beyond diagnosis. In contrast, the proposed framework provides stage-wise hair loss classification, automated image-quality filtering, personalized preventive recommendations, and geolocation-based dermatologist connectivity. This integrated approach aligns with recent calls in digital dermatology research for patient-centric, end-to-end diagnostic support systems rather than isolated prediction models.

Classical machine learning approaches such as KNN and SVM, while computationally simpler, generally exhibit reduced robustness to noise and visual variability in heterogeneous image datasets. The AlphaNet model demonstrates consistently high recall across clinically important classes, minimizing false negatives and making it particularly suitable for early screening applications, where missed detections can delay timely intervention. The near-perfect recall observed for Alopecia Areata and advanced-stage classes further supports its reliability as a first-line diagnostic aid.

Overall, compared with existing state-of-the-art methods, the proposed AlphaNet framework offers a balanced combination of high diagnostic accuracy, low computational overhead, real-world applicability, and functional completeness. Its ability to operate effectively on non-clinical smartphone images, together with mobile deployment and preventive guidance, distinguishes it from prior approaches and supports its suitability for large-scale, accessible hair loss screening and early intervention within modern digital health ecosystems.

## 7. Conclusion and Discussion

This study introduced AlphaNet, a lightweight convolutional neural network integrated into a mobile framework for automated, stage-wise detection and prevention of hair loss from smartphone-captured scalp images. The proposed system addresses important gaps in accessibility and early screening by combining a tailored preprocessing and image-quality filtering pipeline, a compact

CNN architecture optimized for on-device inference, and application-level features that deliver personalized prevention guidance and geolocation-based dermatologist connectivity. Empirical evaluation on the curated dataset (1,402 images) demonstrated strong performance: AlphaNet achieved an overall test accuracy of 98% with a weighted F1-score of 0.98 and high per-class recall for clinically important categories (mostly  $\geq 0.95$ ). These results indicate that a carefully designed, resource-efficient CNN can reliably screen for multiple clinically meaningful stages of hair loss using casual mobile images, making early detection more scalable and widely accessible.

Beyond accuracy, the system’s automatic filtering of low-quality or invalid images proved valuable for improving downstream predictions and user experience: the Invalid class was detected with near-perfect recall, preventing noisy inputs from degrading classification performance. AlphaNet also showed particularly strong discrimination for clinically distinct conditions such as Alopecia Areata, while intermediate stages (notably Stage 2) exhibited lower precision, which highlights the practical challenge of separating adjacent stages from single 2D images. Taken together, the findings support the utility of the approach as a first-line screening and educational tool rather than a substitute for clinical diagnosis.

From an application and impact perspective, the proposed framework advances mobile dermatological screening by enabling low-latency, on-device inference suitable for resource-constrained settings, coupling automated detection with actionable, personalized prevention recommendations, and facilitating pathways to professional care through in-app referral features. These elements collectively increase the potential for earlier user engagement, timely intervention, and broader outreach to populations with limited access to dermatology services.

### 7.1. Limitations

While the proposed Hair Loss Detection and Prevention System shows great promise in automating diagnosis through image-based classification, several limitations need to be recognized to maintain realistic expectations of its current capabilities. First, the system’s diagnostic accuracy relies heavily on the quality, clarity, and angle of the input images. Because it primarily uses 2D scalp photos captured on mobile devices, environmental factors like poor lighting, motion blur, or inconsistent camera focus can significantly affect result reliability. The lack of standardized photo-taking protocols among users also adds variability to the outcomes. Second, the model does not yet account for individual clinical or demographic factors like age, gender, hormone profiles, or genetics all of which are critical in precisely identifying the reasons of hair loss. Its capacity to offer highly customized recommendations is restricted by this deficiency. Furthermore, the training dataset still lacks diversity in terms of hair kinds, ethnic backgrounds, and

scalp conditions, despite being enhanced for robustness. This restriction can affect how broadly the model can be used in practical situations. Lastly, the system is not intended to take the place of expert dermatological assessments. Instead of serving as a conclusive diagnostic platform, it serves as a helpful, first screening tool. It is still not yet possible to integrate it with clinical workflows and long-term health tracking; this is a future objective.

## 7.2. Discussions

To address these limitations and to improve robustness, generalizability, and clinical readiness, we propose the following directions for future work and development:

- **Dataset Expansion and Diversity:** Collect and annotate substantially larger, more diverse datasets with balanced representation across hair types, skin tones, ages, and geographic populations. Prioritize additional examples for underrepresented classes (e.g., Stage 2) to improve per-class precision.
- **Multimodal and Clinical Integration:** Incorporate clinical metadata (patient history, medication use, hormone profiles) and multimodal imaging (trichoscopic/dermoscopic photos, OCT) to enable richer feature representations and improve discrimination between adjacent stages and etiologies.
- **Temporal Monitoring and Longitudinal Models:** Extend the system to support longitudinal tracking (time-series of user-submitted images) and develop models that exploit temporal changes to increase sensitivity for early progression and treatment response monitoring.
- **Explainability and Trust:** Integrate explainability techniques (saliency maps, class activation mapping) and human-centered explanations to make model outputs more interpretable for users and clinicians, thereby improving trust and adoption.
- **On-Device Optimization and Privacy-Preserving Learning:** Further optimize the model for low-power

devices (quantization, pruning) and explore federated or differential-privacy training strategies to preserve user privacy while benefiting from distributed data.

- **Active Learning and Continuous Improvement:** Deploy active learning pipelines where uncertain or misclassified cases are flagged for expert review and incremental retraining, improving model robustness over time with minimal labeling overhead.
- **Clinical Validation and Usability Studies:** Conduct prospective clinical trials and usability studies to evaluate diagnostic concordance with dermatologists, the system’s impact on care-seeking behavior, and real-world acceptance across diverse user groups.
- **Regulatory and Ethical Readiness:** Initiate early engagement with regulatory frameworks and ethics boards to ensure the application meets medical-device standards, informed-consent practices, and transparent risk-communication policies before large-scale deployment.
- **Feature Expansion:** Explore integration with wearable/hair-sensor data, teledermatology workflows, and treatment-adherence modules to create a more comprehensive hair-health management ecosystem.

## 7.3. Concluding Remarks

In summary, AlphaNet demonstrates that lightweight, mobile-optimized CNN architectures can deliver high-performance, stage-wise hair loss screening from smartphone images while enabling user-facing prevention and referral features. By combining technical refinements (dataset expansion, multimodality, temporal modeling) with rigorous clinical validation and strong privacy safeguards, the system can evolve into a practical, ethically responsible tool to support earlier detection and improved access to dermatological care. The directions outlined above chart a realistic path toward increasing clinical utility, robustness, and adoption in real-world settings.

## Declarations

**Ethics approval and consent to participate:** Ethical approval is not required, and the authors affirm their voluntary involvement in the development of the work.

**Consent for publication:** Written consent has been obtained from each researcher to publish their data and contributions in the present research paper.

**Availability of data and material:** Not Applicable

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